UCT41042 – Practical for Introduction to Smart System

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Faculty of Technology

South Eastern University of Sri Lanka

Lab Sheet No: 02

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Title: Over-The-Air (OTA) Firmware Updates with ESP32

Aims: To introduce the concept, importance, and implementation of Over-The-Air (OTA) updates on the ESP32 platform. Students will learn how to enable wireless firmware updates, perform updates remotely and handle OTA events in IoT projects.

Tasks:

- Understand the OTA process and its benefits (remote maintenance, no physical access needed).
- Implement OTA using the ArduinoOTA library.
- Monitor and handle OTA events (start, progress, end, errors).
- Perform version updates and verify success.
- Explore visual feedback.

Key Concepts

- ❖ OTA: Firmware update delivered wirelessly (usually over Wi-Fi).
- ❖ Partitions: ESP32 uses separate OTA partitions to safely swap between old and new firmware.
- ❖ Workflow: Device connects to Wi-Fi → Listens for update → Downloads new binary → Validates → Reboots into new firmware.
- ❖ Advantages: Easy field updates, scalability for multiple devices.
- ❖ Risks: Interrupted updates, incompatible firmware, security vulnerabilities.

Exercise 1: Basic OTA Setup with ArduinoOTA

1. Create a new ESP32 project in Wokwi or open in Arduino IDE.
2. Connect a LED with a 220Ω current-limiting resistor to GPIO 2 (Pin 2) of the ESP32.
3. Write a sketch that connects to Wi-Fi and enables ArduinoOTA.
4. Include proper error handling and event callbacks.
5. Upload the initial version (call it Version 1) and observe the Serial Monitor.

Expected Output in Serial Monitor:

Booting message
Wi-Fi connection status
IP address
OTA ready confirmation

Exercise 2: Performing Your First OTA Update

1. Modify the initial sketch to create Version 2:
   - Change the version message.
   - Modify LED blink pattern (e.g., faster blink or different duty cycle).
   - Add a new feature (e.g., print "OTA Update Successful – Now Running v2").
2. In Arduino IDE:
   - Select the network port (ESP32's IP address).
   - Upload the new sketch over-the-air.
3. Observe the update progress, completion message, and automatic reboot.
4. Verify the new behavior (LED pattern + Serial output).